

PRAKRIT RAMACHANDRAN

Mechanical Engineer — Design, CAD & Motion Systems

prakrit@uw.edu • prakrit-ramachandran.netlify.app • github.com/prakritr

SUMMARY

Mechanical engineering student focused on design and motion systems, with hands-on experience owning builds from CAD, FEA, and closed-loop control through fabrication and validation. Skilled in SolidWorks, Fusion 360, and AutoCAD, producing detailed assemblies and manufacturing drawings while iterating designs against weight, structural, and performance constraints.

EDUCATION

University of Washington — B.S. Mechanical Engineering

Expected 2027

Seattle, WA

2024 – Present

Relevant Coursework: System Dynamics & Controls, Kinematics, Statics, Mechanics of Materials, Materials Science, Thermodynamics, Manufacturing Processes, CAD & Engineering Graphics

EXPERIENCE

Mechanical Engineer — Husky Robotics, Drone Subteam

Collegiate robotics team

Seattle, WA

Jan 2025 – Jan 2026

- Led mechanical design and fabrication of a competition drone through a 10-week build cycle, meeting payload, weight, and structural constraints; produced detailed CAD assemblies and manufacturing drawings.
- Ran FEA in SolidWorks to optimize airframe geometry for weight and flight efficiency while preserving structural integrity under load.
- Designed and prototyped a multi-orientation gripping end-effector for reliable object capture during dynamic flight, iterating through rapid prototyping and flight testing.
- Integrated mechanical systems with the electrical and software subteams and defended designs in formal design reviews.

Co-Founder & Software Lead — Axis

AI-powered property management platform

Seattle, WA

2026 – Present

- Co-founded and led a 3-person team to build an AI-powered property management platform from the ground up, shipping web and iOS apps to production (React Native/Expo, Next.js, Supabase, Stripe).
- Architected and integrated multiple third-party systems — payments, tenant screening, and SMS — into one full-stack product, owning the build end-to-end across frontend, backend, and infrastructure.
- Ran go-to-market and customer acquisition through direct cold-calling and outreach to landlords and property managers, onboarding the platform's first 2 landlords and 28 rooms.

Facilities Engineer Intern — Quanta Labs

Test & measurement hardware

Seattle, WA

Jun 2023 – Aug 2023

- Designed 4+ AutoCAD schematics for shock and vibration test systems, supporting manufacturing and deployment timelines.
- Contributed to PCB layout for vibration test systems, improving measurement accuracy and data consistency.

Team Captain — FIRST Tech Challenge

Competitive robotics

Fremont, CA

Aug 2018 – Mar 2024

- Recruited and led an 11-member engineering team across three functional subteams while managing a \$10K annual budget.
- Directed mechanical architecture, detailed design, and assembly of six 500+ part competition robots under strict seasonal deadlines.
- Integrated closed-loop control using PID, gyroscope, and odometry feedback for autonomous navigation and precision manipulation; competed at Regional, World Championship, and Chicago Invitational levels, earning 30+ team awards.

PROJECTS

2-Axis Pen Plotter — CoreXY Motion System

- Designed and built a precision 2-axis CoreXY motion system using belt-driven actuation and stepper motors; engineered 20+ custom components and tuned frame rigidity and belt tension to reduce vibration and improve positional accuracy.
- Implemented an Arduino-based control system running GRBL firmware for synchronized motion, calibrated for repeatability through iterative testing.

Marble Coaster — Continuous-Motion Mechanism

- Designed and fabricated a continuous-motion mechanical system from concept to assembly, modeling 10+ custom components and applying kinematics and energy-conservation principles to optimize track curvature and eliminate derailment.

SKILLS

CAD & Analysis: SolidWorks, Fusion 360, Onshape, AutoCAD, CAD Assemblies, GD&T (basic), Mechanical Simulation, Design for Manufacturing

Robotics & Mechatronics: Closed-Loop / PID Control, Motion Control Systems, Stepper/Servo Actuation, GRBL, Sensor Feedback (Gyro, Odometry), Systems Integration

Prototyping & Fabrication: 3D Printing (FDM), CNC Machining, Laser Cutting, Welding, Rapid Prototyping

Electronics & Embedded: Arduino, Raspberry Pi, PCB Layout (basic), Embedded Motion Control

Programming: Python, TypeScript / JavaScript, Java, MATLAB, SQL

Full-Stack & AI Development: React Native / Expo, Next.js, Node.js, REST APIs, Supabase (PostgreSQL), Stripe, Git; agentic / AI-assisted coding with LLM APIs and prompt engineering

Business & Go-to-Market: Customer Acquisition, Cold Calling & Outreach, Property Management, Market Research